Operating Systems: Internals and Design Principles

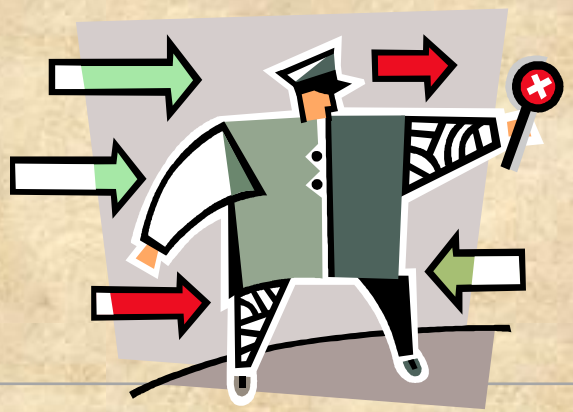
Chapter 1 Computer System Overview

Objectives

- To describe the basic organization of computer systems
- To provide a grand tour of the major components of operating systems
- To give an overview of the many types of computing environments
- To explore several open-source operating systems

Operating System

- Controls the execution of application programs and acts as an interface between the user of a computer and the computer hardware.
- Provides a set of services to system users
- Manages secondary memory and I/O devices



Functions of Operating System

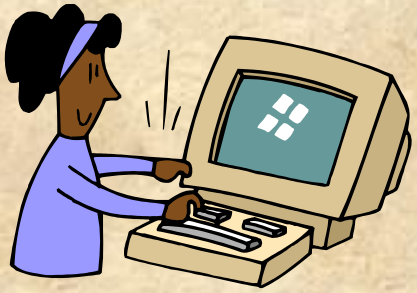
- Processor Management
- Memory Management
- Device Management
- File Management
- Security
- Job scheduling

Types of Operating System

- Batch Operating system
- Multi-programming OS
- Multi-tasking OS
- Multi-processing OS
- Real-time OS
- Distributed OS

Basic Elements

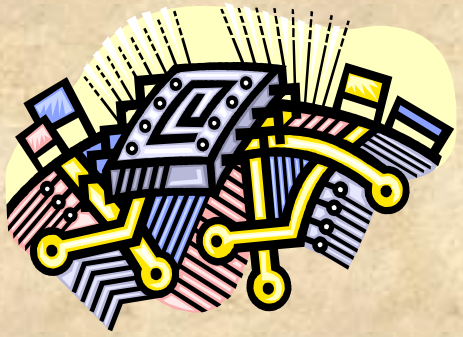
- Processor
- I/O Modules
- Main Memory
- System Bus



Processor

Controls the operation of the computer

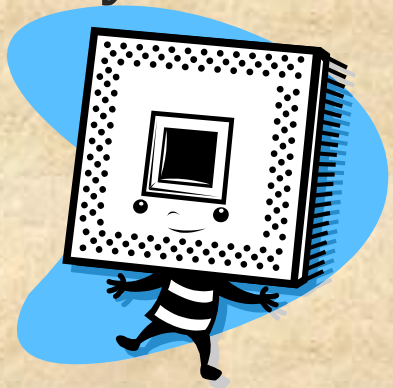
Performs the data processing functions



Referred to as the *Central Processing Unit* (CPU)

Main Memory

- Volatile
- Contents of the memory is lost when the computer is shut down
- Referred to as real memory or primary memory



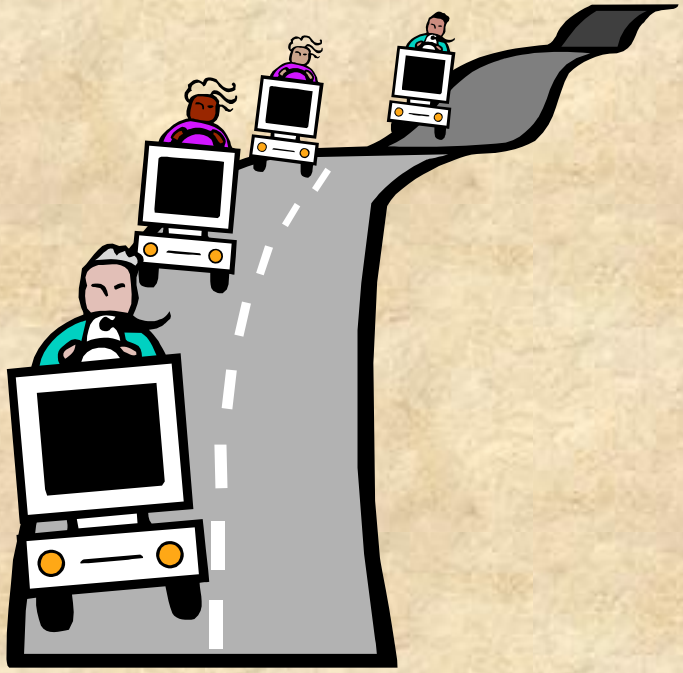
I/O Modules

Moves data between the computer and external environments such as:

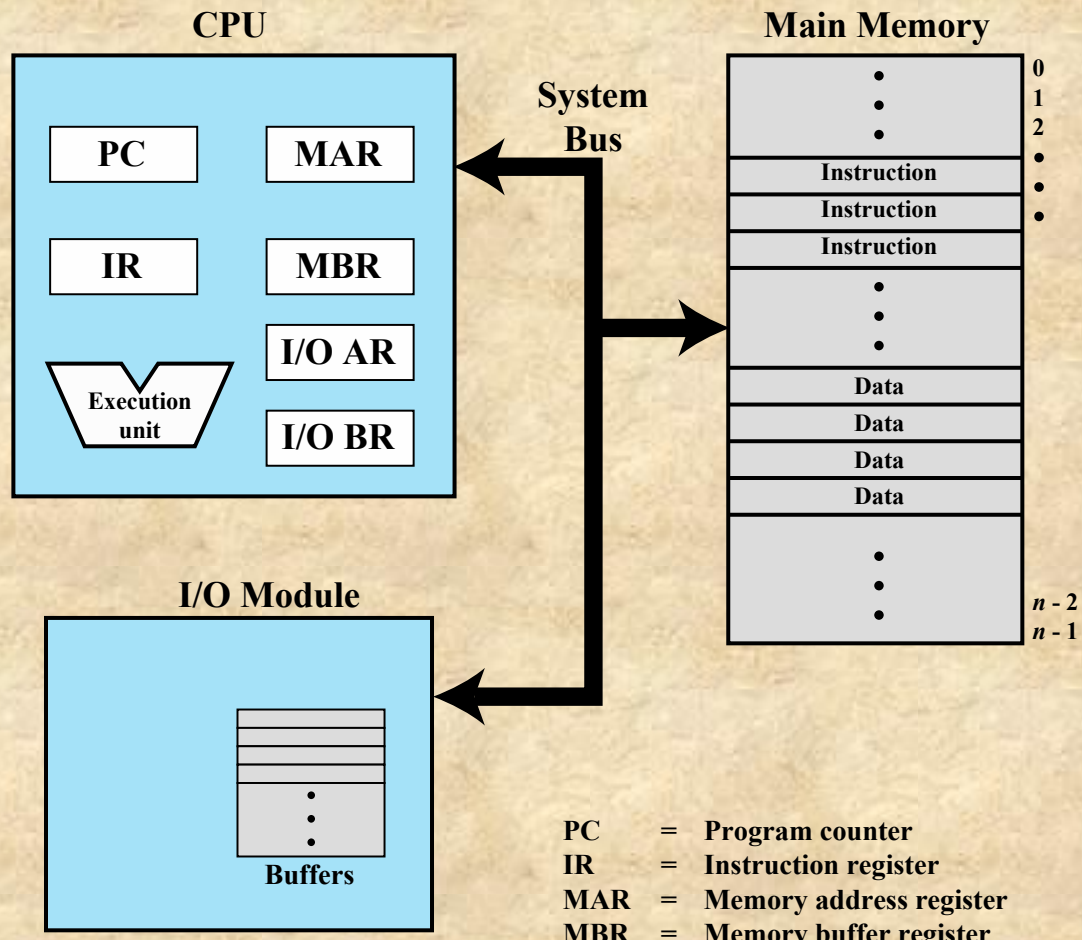
- storage (e.g. hard drive)
- communications equipment
- terminals

System Bus

- Provides for communication among processors, main memory, and I/O modules

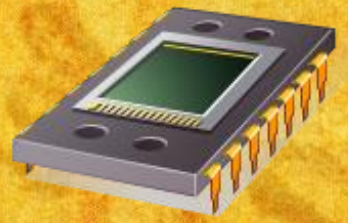


Computer components : Top-level view



PC = Program counter
IR = Instruction register
MAR = Memory address register
MBR = Memory buffer register
I/O AR = Input/output address register
I/O BR = Input/output buffer register

Microprocessor



- Invention that brought about desktop and handheld computing
- Processor on a single chip
- Fastest general-purpose processor
- Multiprocessors
- Each chip (socket) contains multiple processors (cores)

Graphical Processing Units (GPU's)

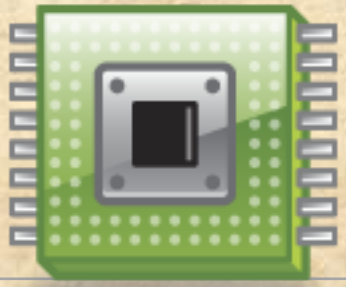
- Provide efficient computation on arrays of data using Single-Instruction Multiple Data (SIMD) techniques
- Used for general numerical processing
- Physics simulations for games
- Computations on large spreadsheets
- Helps to handle graphics-related work like graphics, effects, and videos.

Digital Signal Processors (DSPs)

- Deal with streaming signals such as audio or video
- Used to be embedded in devices like modems
- Encoding/decoding speech and video (codecs)
- Support for encryption and security
- It takes real-world signals like voice, audio, video, temperature, pressure, or position that have been digitized and then mathematically manipulate them.

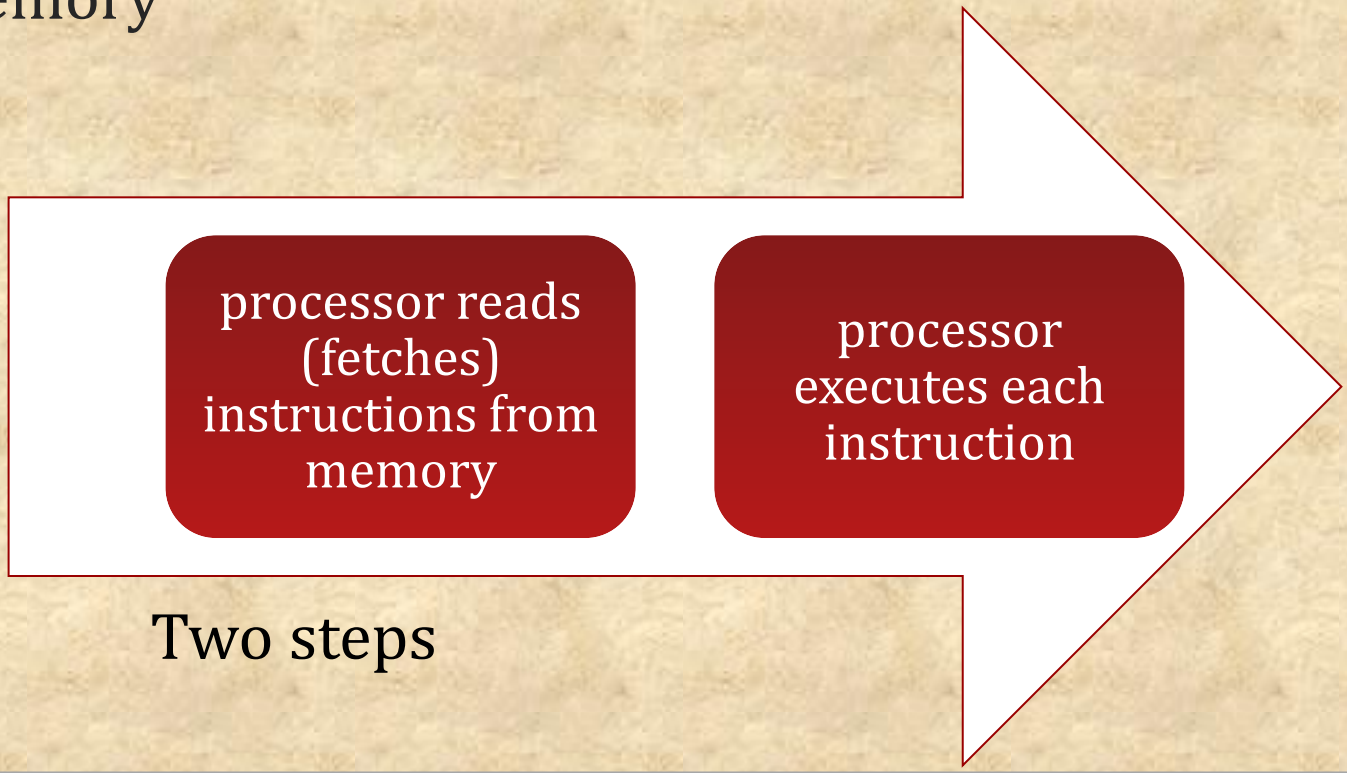
System on a Chip (SoC)

- It is an integrated chip or circuit that combines all the various components of a computer or other system.
- To satisfy the requirements of handheld devices, the microprocessor is giving way to the SoC
- Components such as DSPs, GPUs, codecs and main memory, in addition to the CPUs and caches, are on the same chip



Instruction Execution

- A program consists of a set of instructions stored in memory



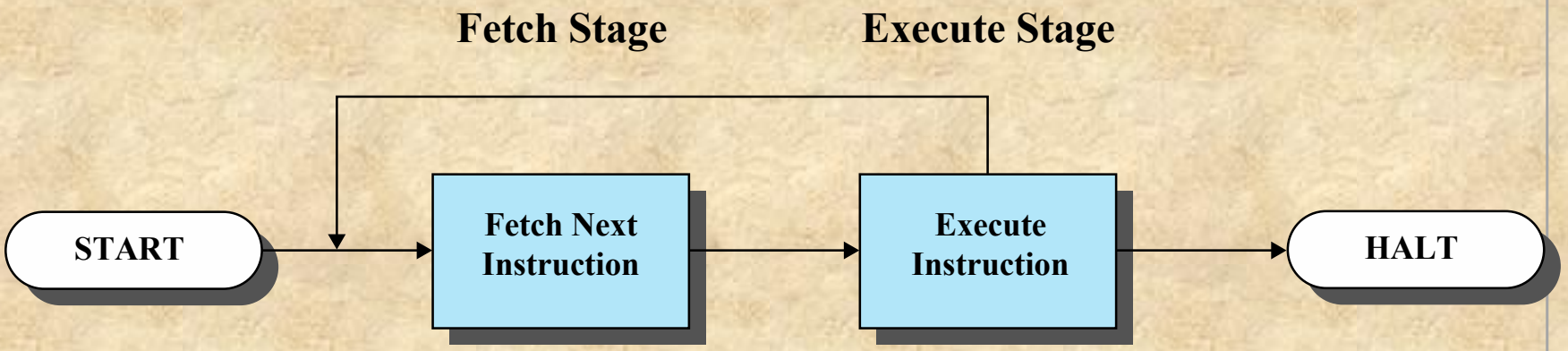
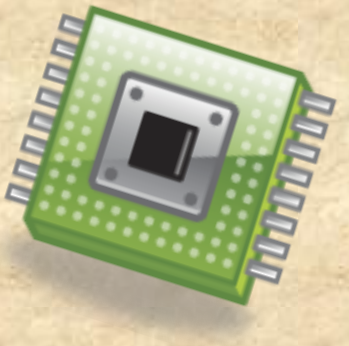


Figure 1.2 Basic Instruction Cycle

Instruction Fetch and Execute

- The processor fetches the instruction from memory
- Program counter (PC) holds address of the instruction to be fetched next
    - PC is incremented after each fetch



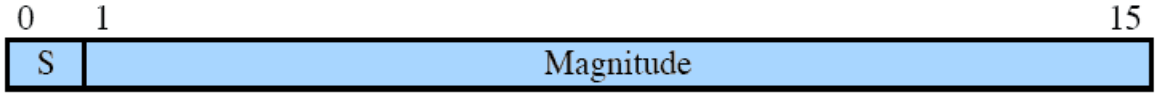
Instruction Register (IR)



- Processor interprets the instruction and performs required action:
  - Processor-memory
  - Processor-I/O
  - Data processing
  - Control



(a) Instruction format



(b) Integer format

Program counter (PC) = Address of instruction
Instruction register (IR) = Instruction being executed
Accumulator (AC) = Temporary storage

(c) Internal CPU registers

0001 = Load AC from memory
0010 = Store AC to memory
0101 = Add to AC from memory

(d) Partial list of opcodes

Figure 1.3 Characteristics of a Hypothetical Machine

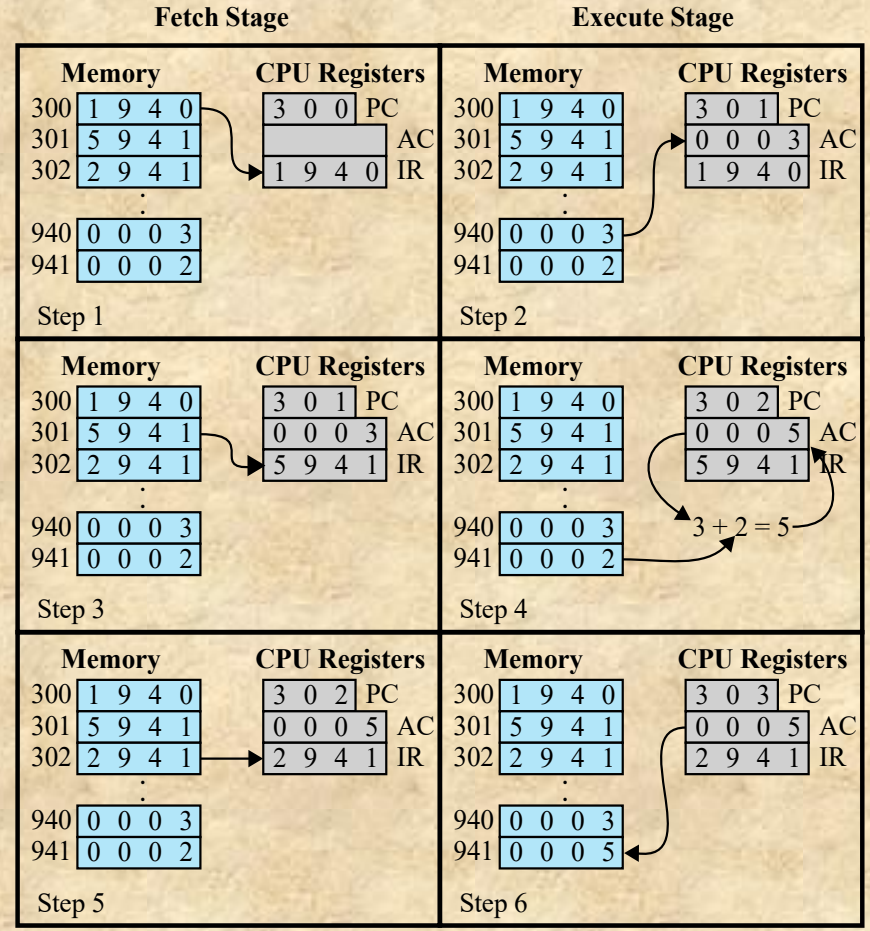


Figure 1.4 Example of Program Execution
(contents of memory and registers in hexadecimal)

Interrupts

- Interrupt the normal sequencing of the processor
- Provided to improve processor utilization
  - most I/O devices are slower than the processor
  - processor must pause to wait for device
  - wasteful use of the processor

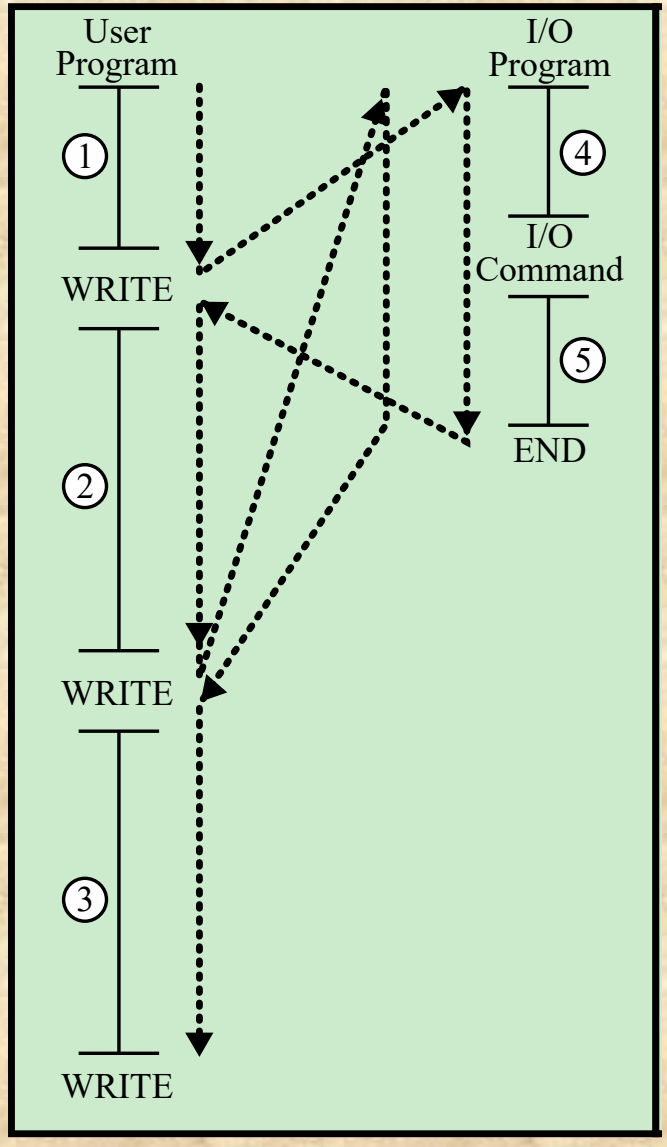


Table 1.1 Classes of Interrupts

Program result of an division	Generated by some condition that occurs as a instruction execution, such as arithmetic overflow, by zero, attempt to execute an illegal machine instruction, and reference outside a user's allowed memory space.
Timer	Generated by a timer within the processor. This allows the operating system to perform certain functions on a regular basis.
I/O	Generated by an I/O controller, to signal normal completion of an operation or to signal a variety of error conditions.
Hardware failure	Generated by a failure, such as power failure or memory parity error.

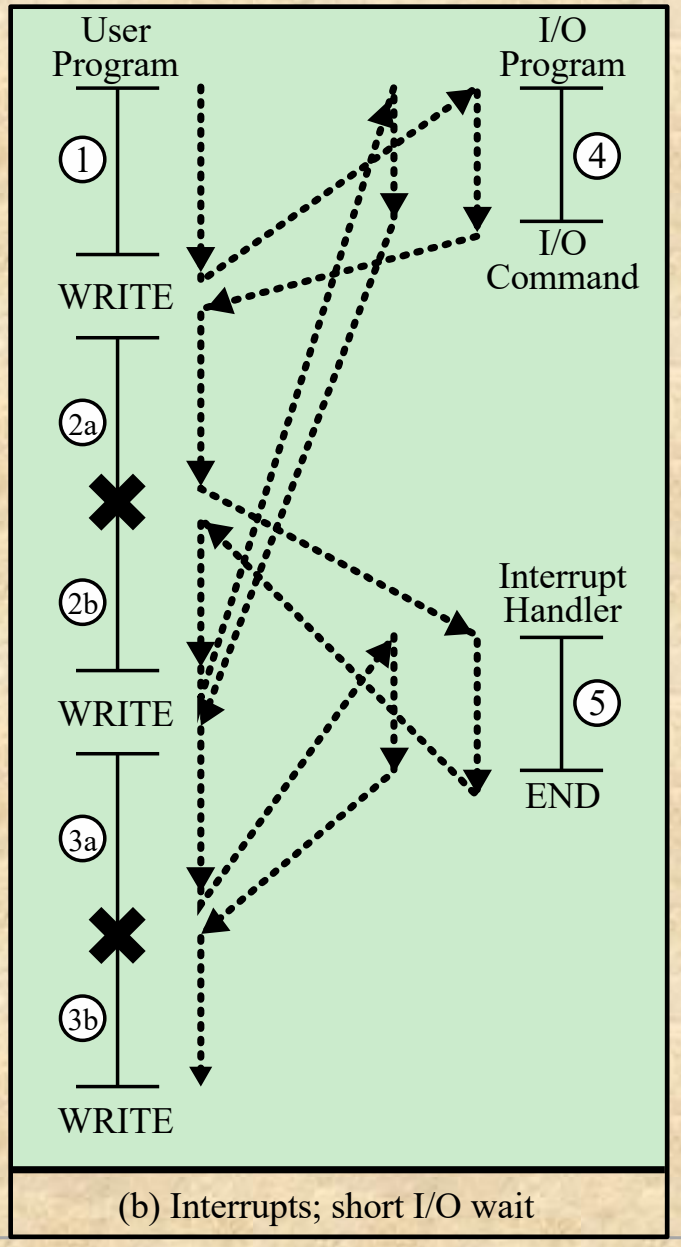
Flow of Control

Without Interrupts



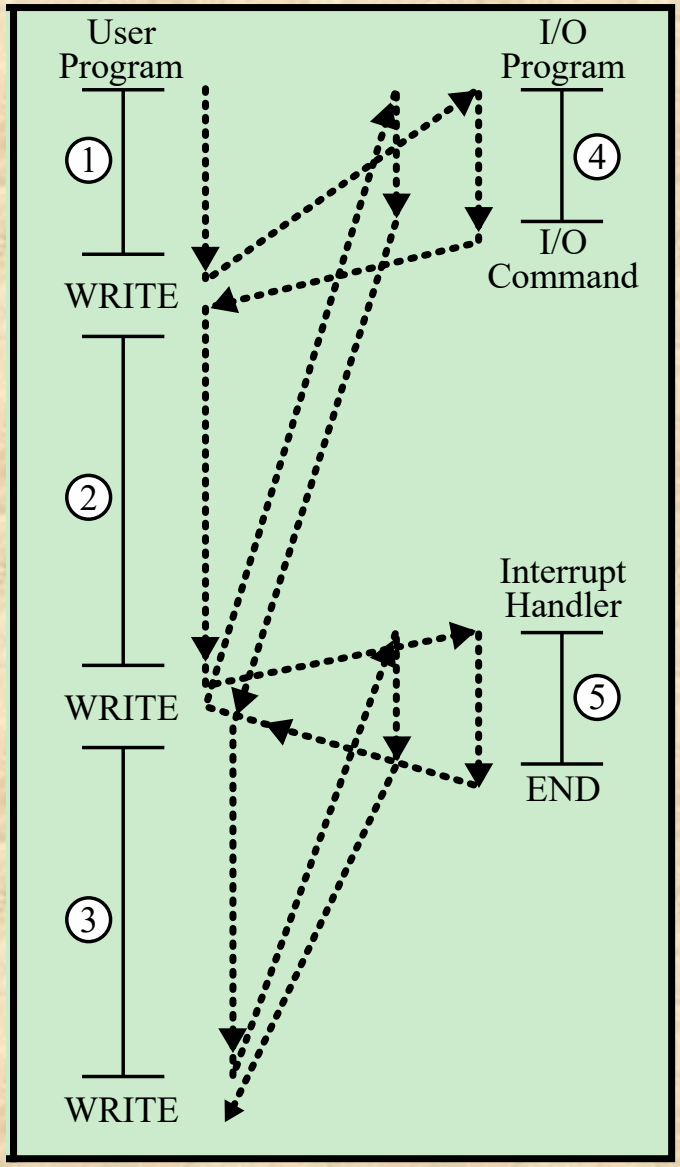
(a) No interrupts

Short I/O Wait



(b) Interrupts; short I/O wait

Long I/O Wait



(c) Interrupts; long I/O wait

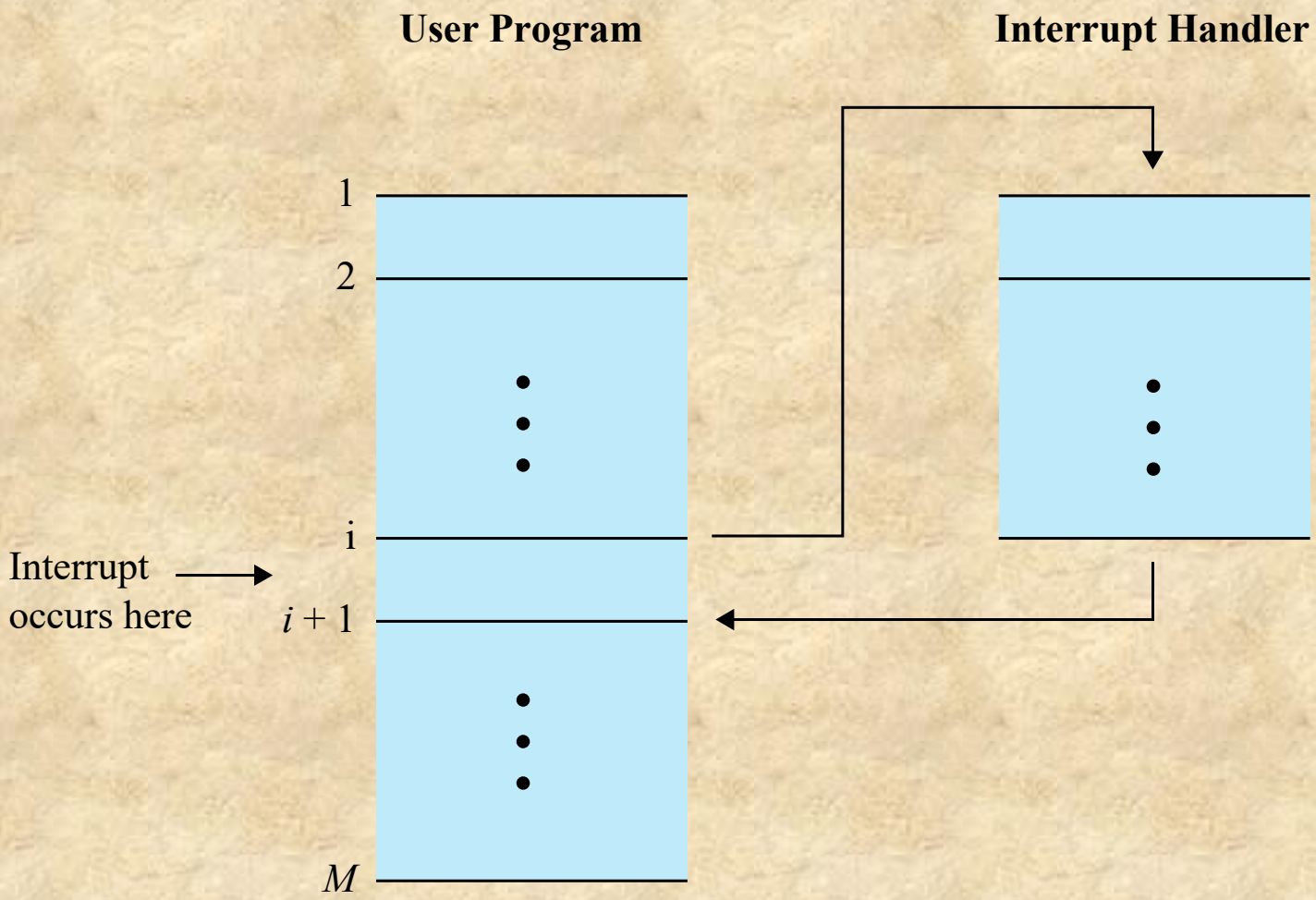


Figure 1.6 Transfer of Control via Interrupts

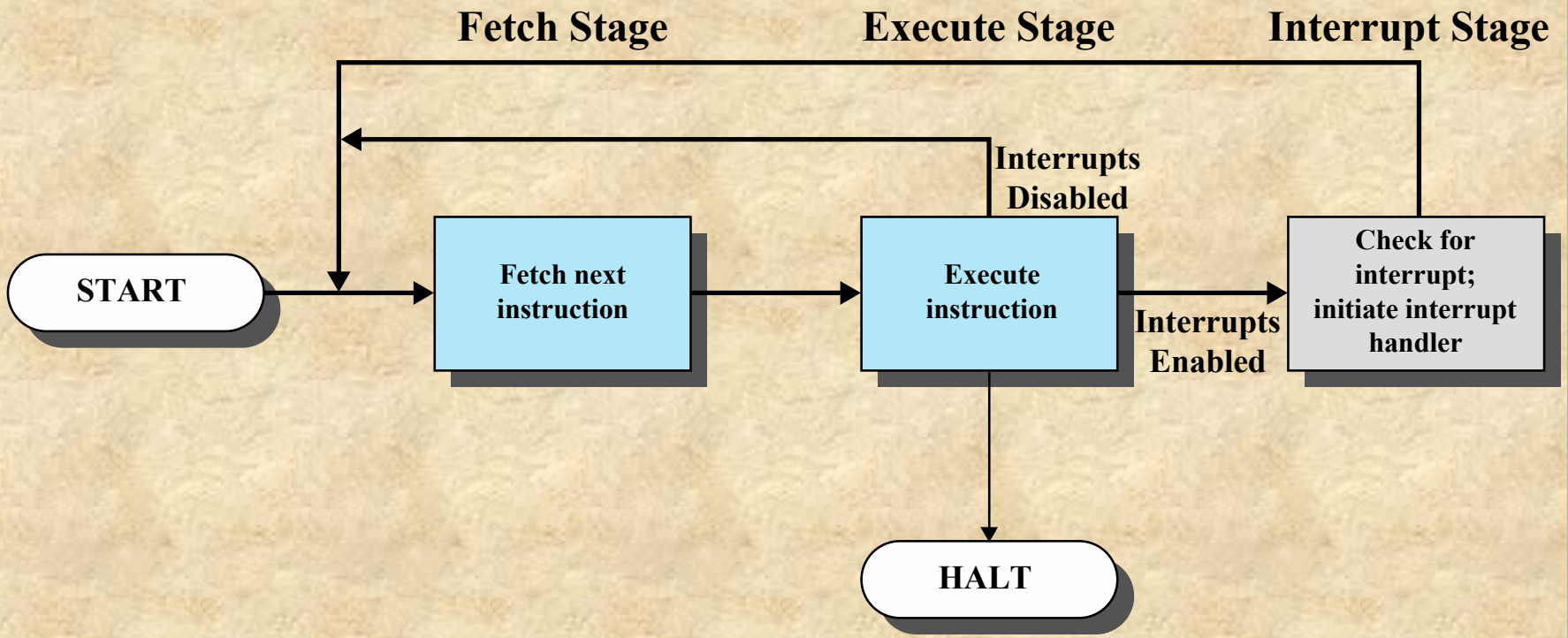


Figure 1.7 Instruction Cycle with Interrupts

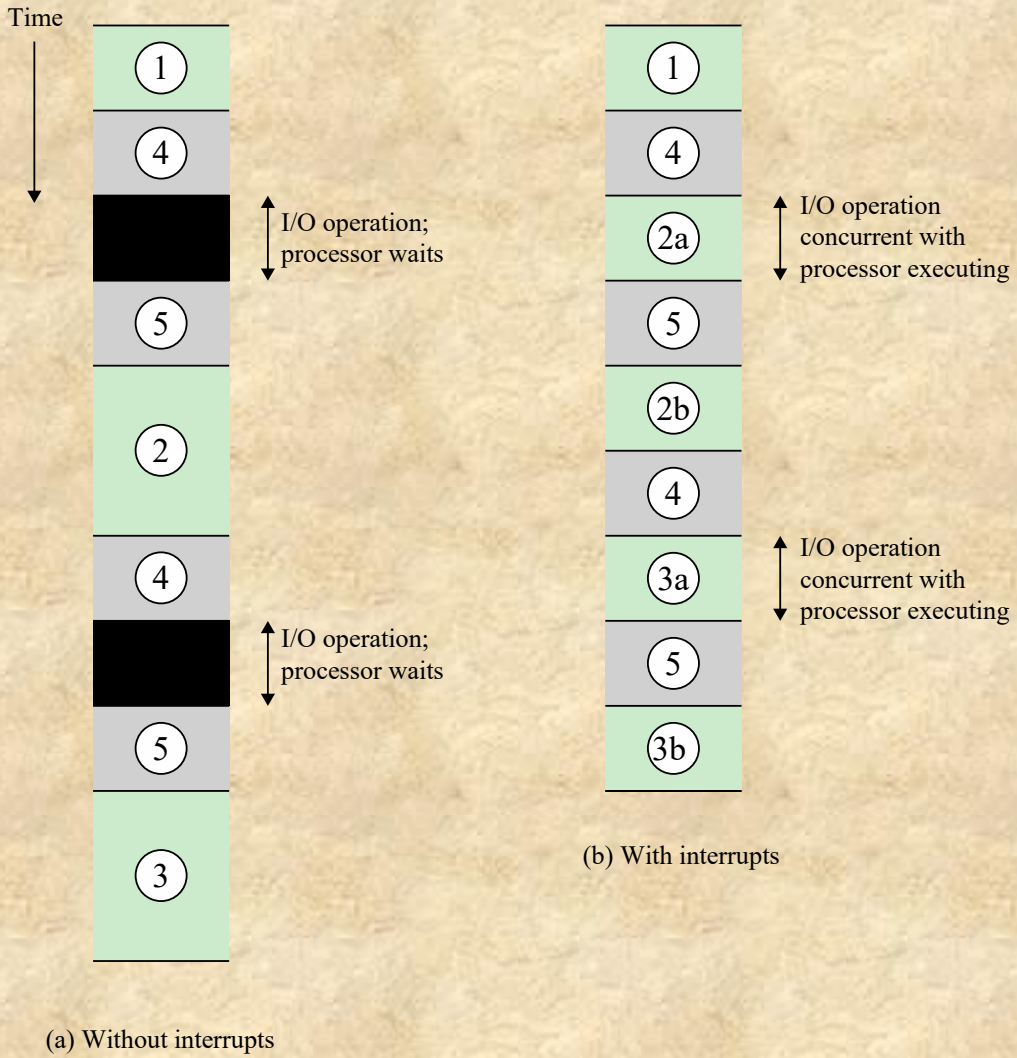


Figure 1.8 Program Timing: Short I/O Wait

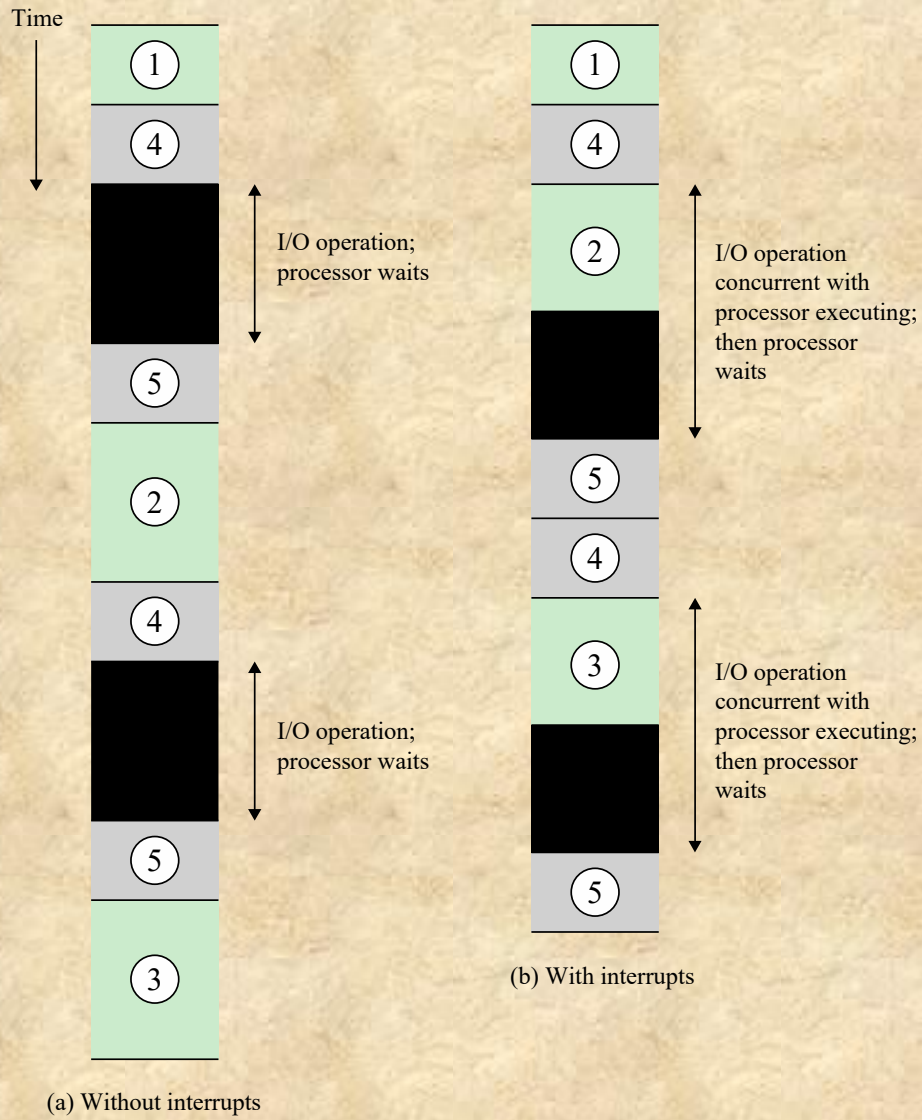


Figure 1.9 Program Timing: Long I/O Wait

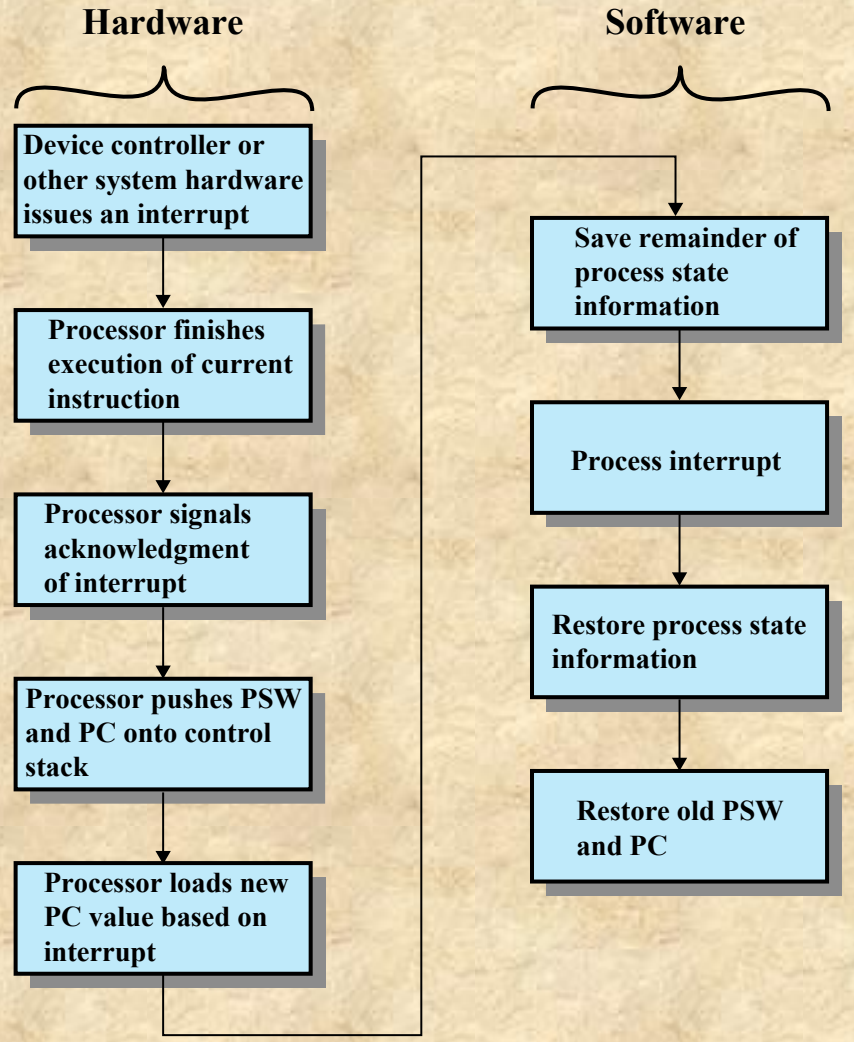


Figure 1.10 Simple Interrupt Processing

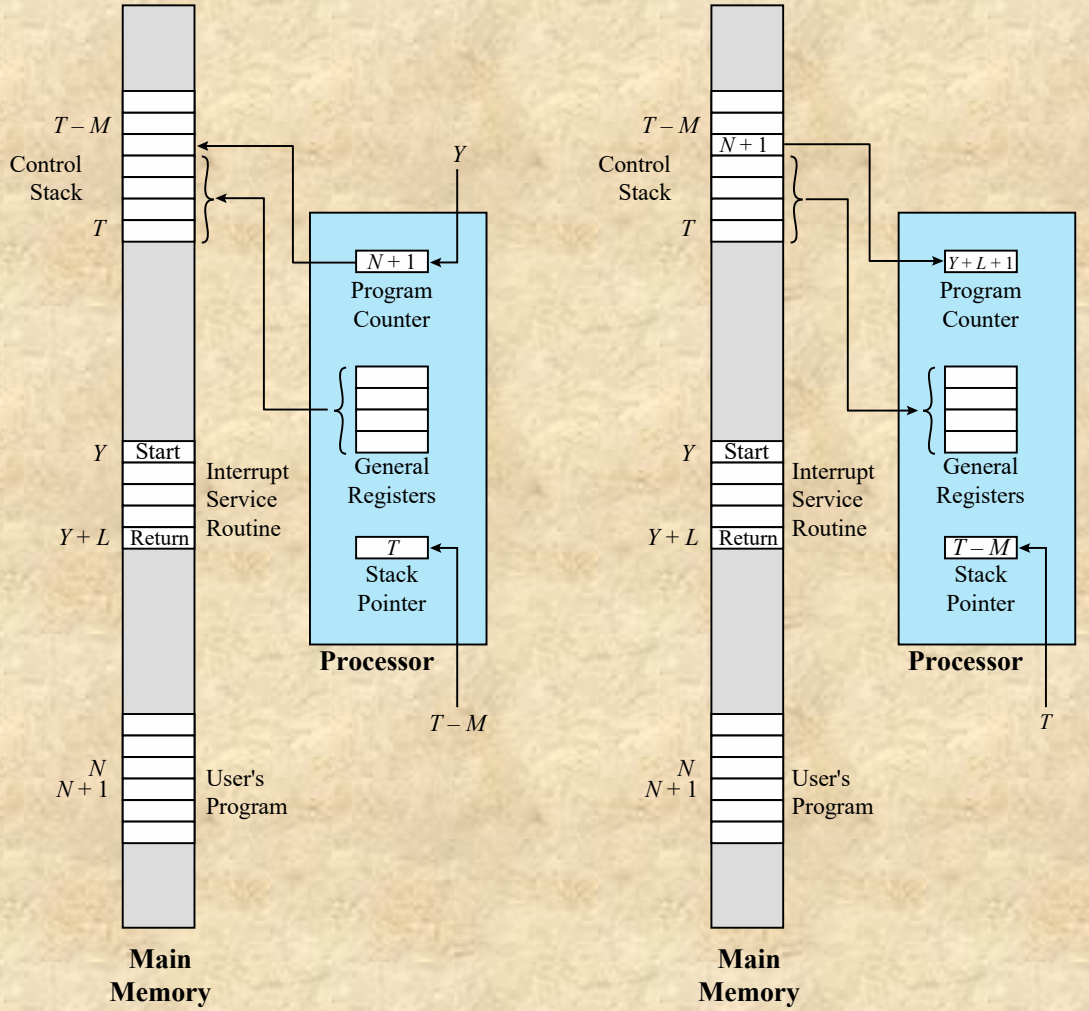


Figure 1.11 Changes in Memory and Registers for an Interrupt

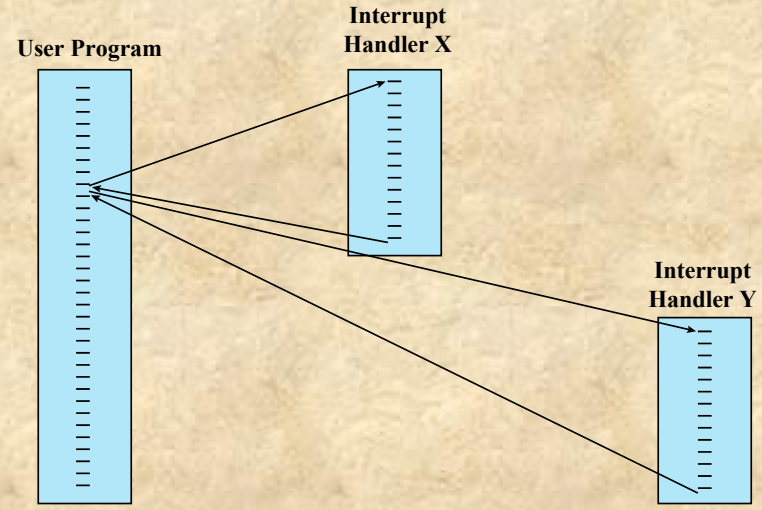
Multiple Interrupts

An interrupt occurs while another interrupt is being processed

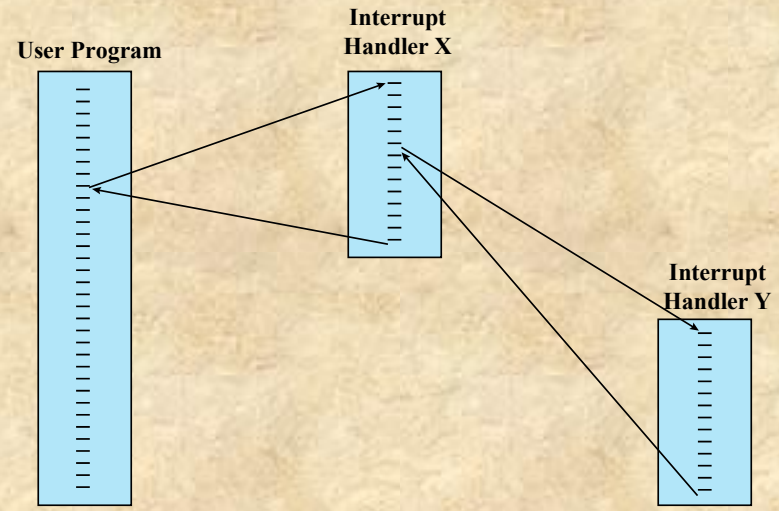
- e.g. receiving data from a communications line and printing results at the same time

Two approaches:

- disable interrupts while an interrupt is being processed
- use a priority scheme



(a) Sequential interrupt processing



(b) Nested interrupt processing

Figure 1.12 Transfer of Control with Multiple Interrupts

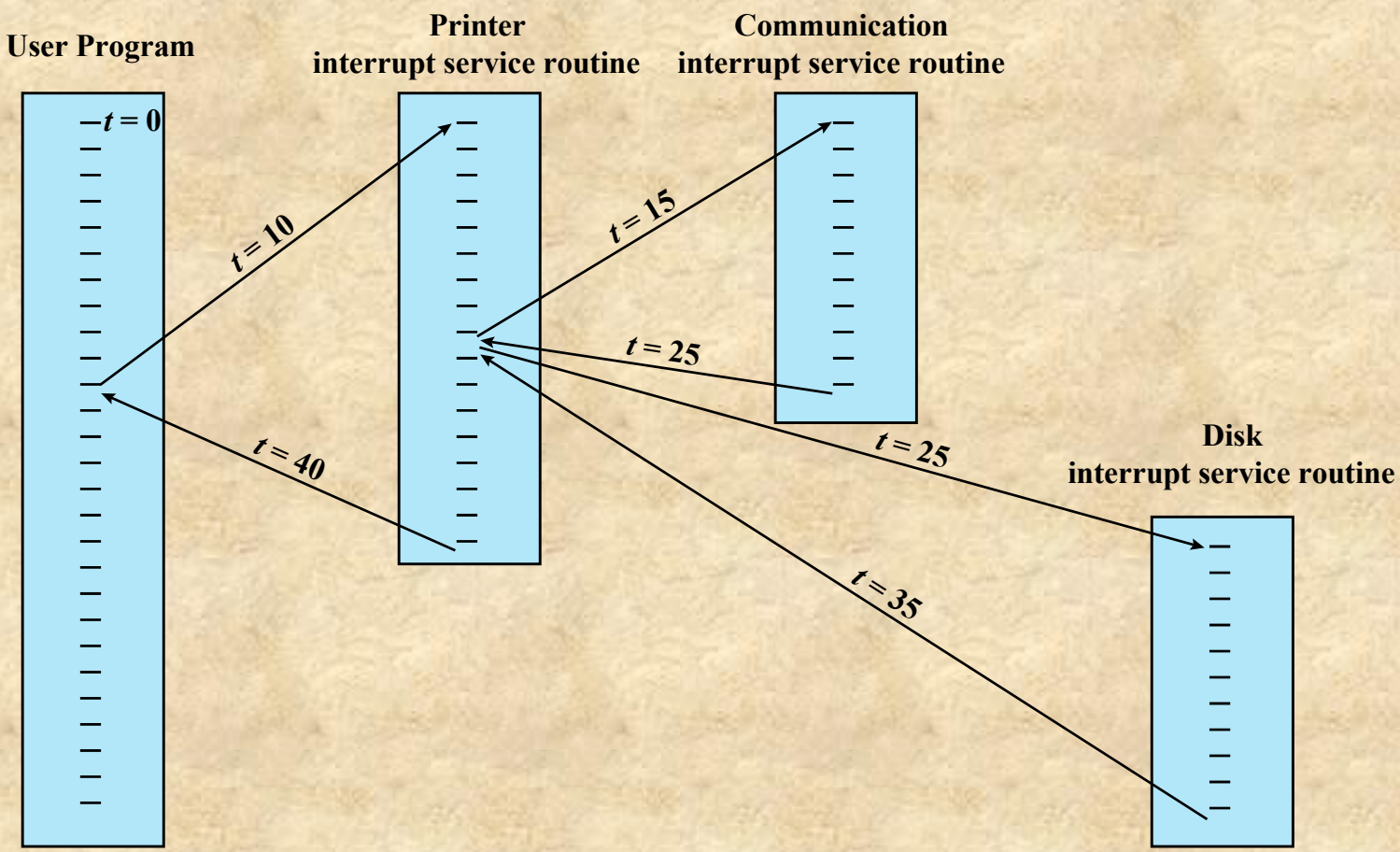
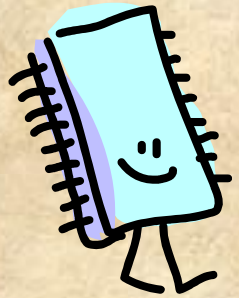


Figure 1.13 Example Time Sequence of Multiple Interrupts

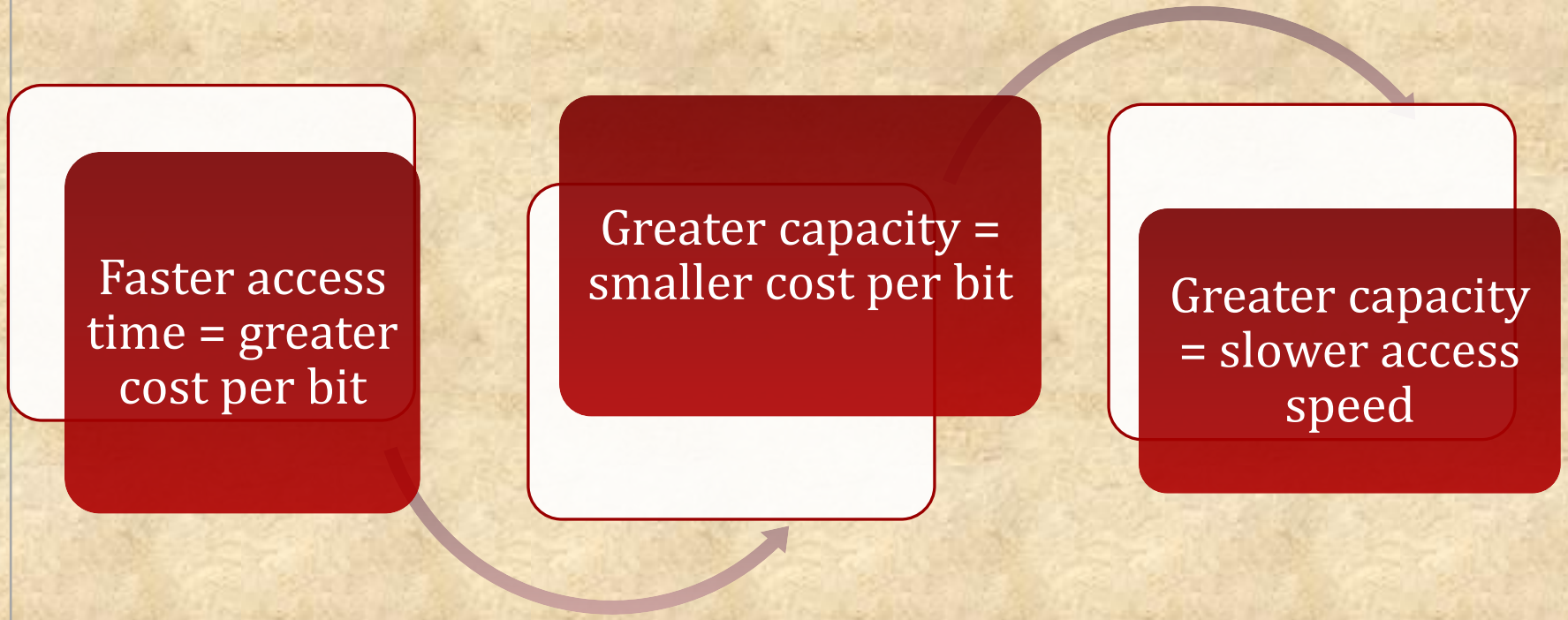
Memory Hierarchy

- Major constraints in memory
  - amount
  - speed
  - expense



- Memory must be able to keep up with the processor
- Cost of memory must be reasonable in relationship to the other components

Memory Relationships



The Memory Hierarchy

- Going down the hierarchy:
  - decreasing cost per bit
  - increasing capacity
  - increasing access time
  - decreasing frequency of access to the memory by the processor

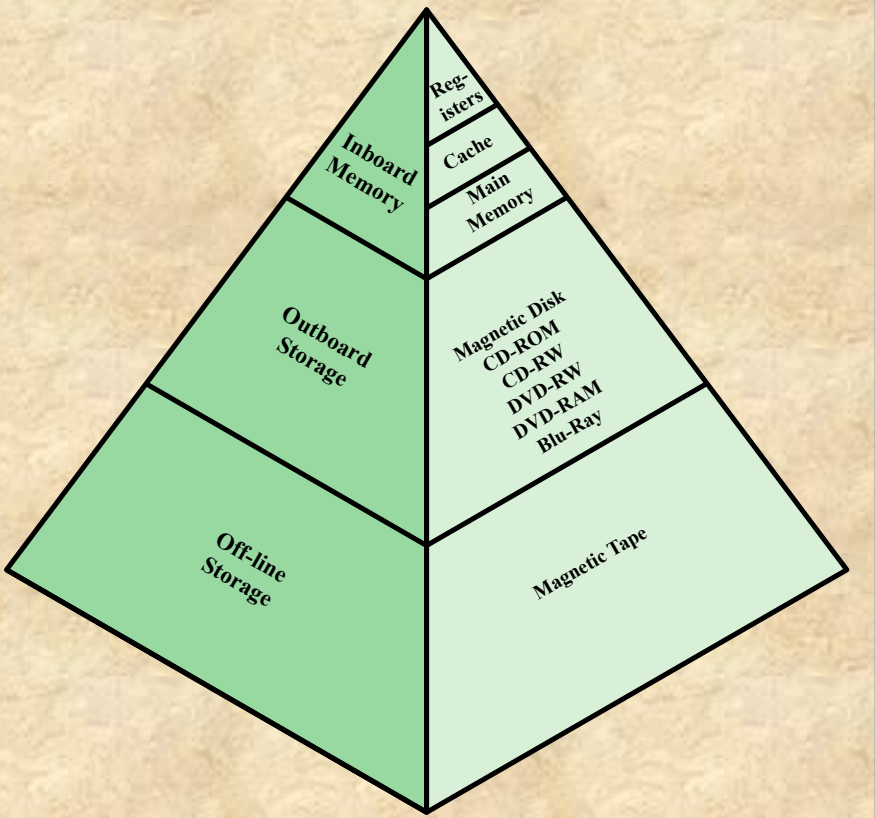


Figure 1.14 The Memory Hierarchy

Principle of Locality

- Also known as Locality of reference
- refers to the tendency of the computer program to access the same set of memory locations for a particular time period.
- The property of Locality of Reference is mainly shown by loops and subroutine calls in a program.

Secondary Memory

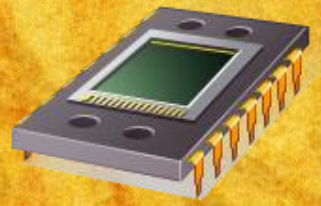




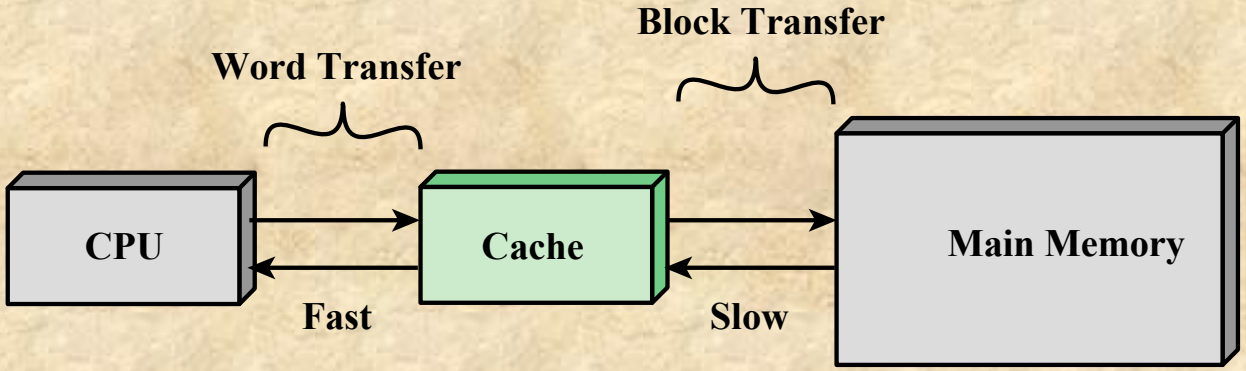
Also referred to as auxiliary memory

- external
- nonvolatile
- used to store program and data files

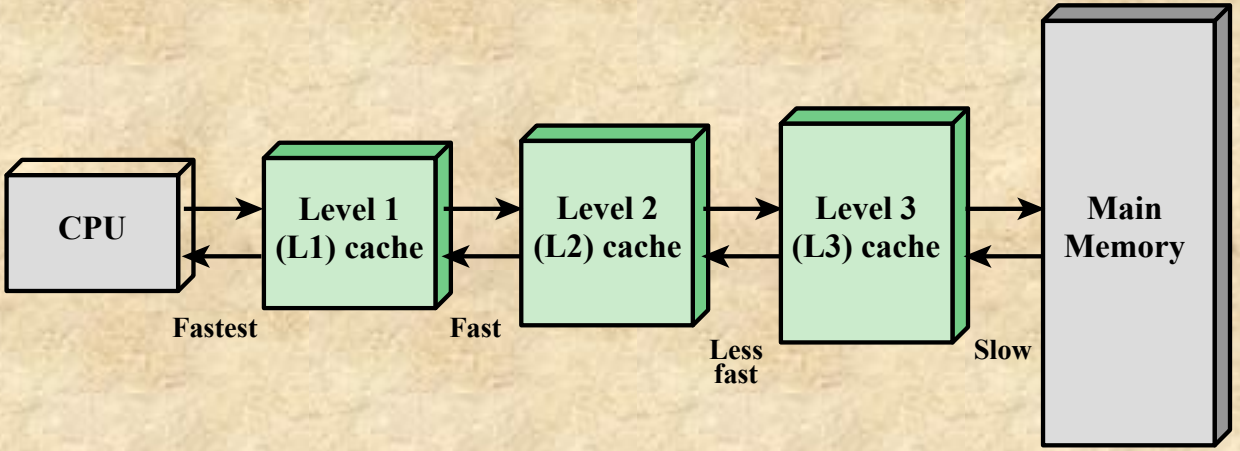
Cache Memory



- Invisible to the OS
- Interacts with other memory management hardware
- Processor must access memory at least once per instruction cycle
- Processor execution is limited by memory cycle time
- Exploit the principle of locality with a small, fast memory



(a) Single cache



(b) Three-level cache organization

Figure 1.16 Cache and Main Memory

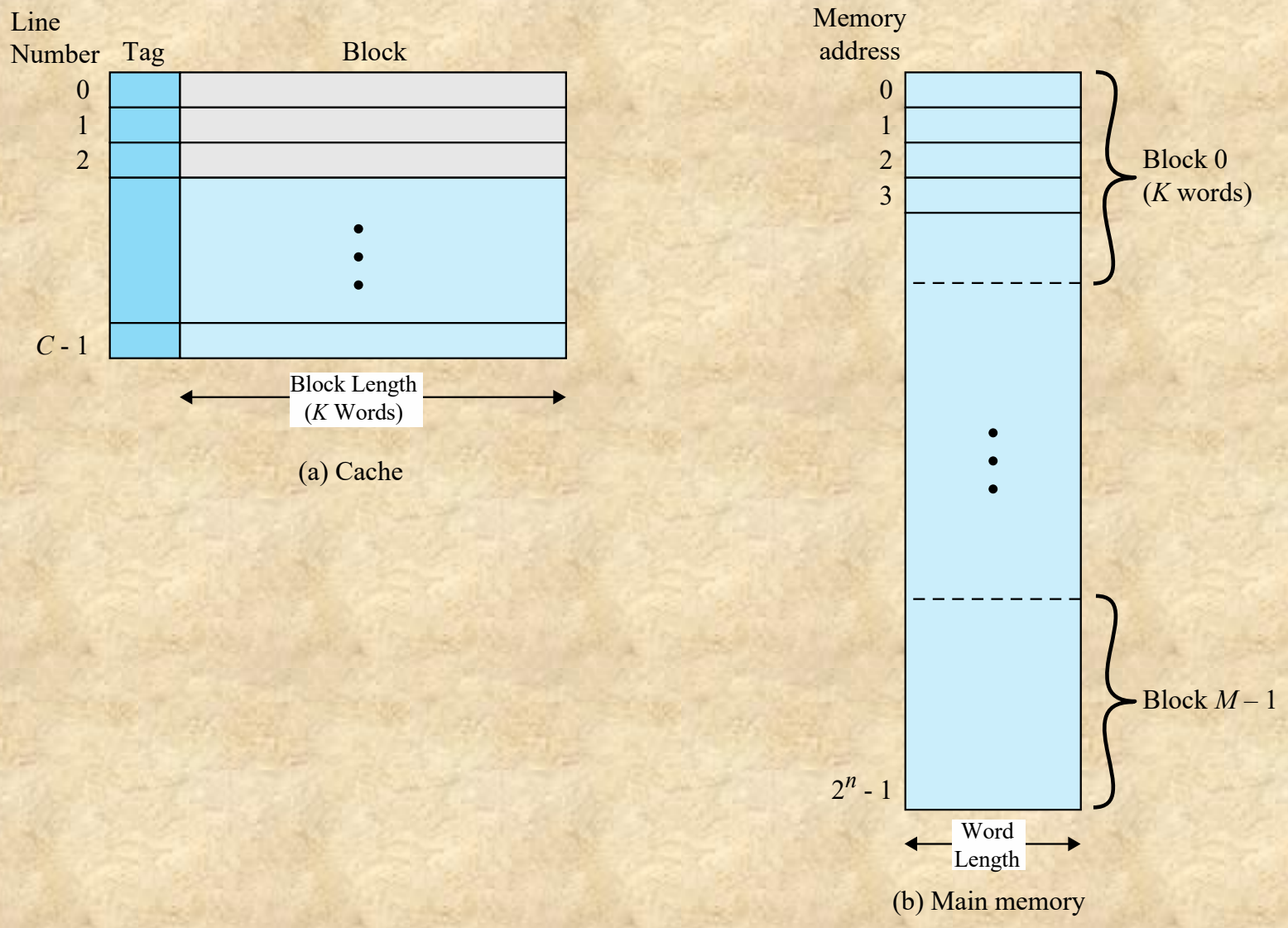


Figure 1.17 Cache/Main-Memory Structure

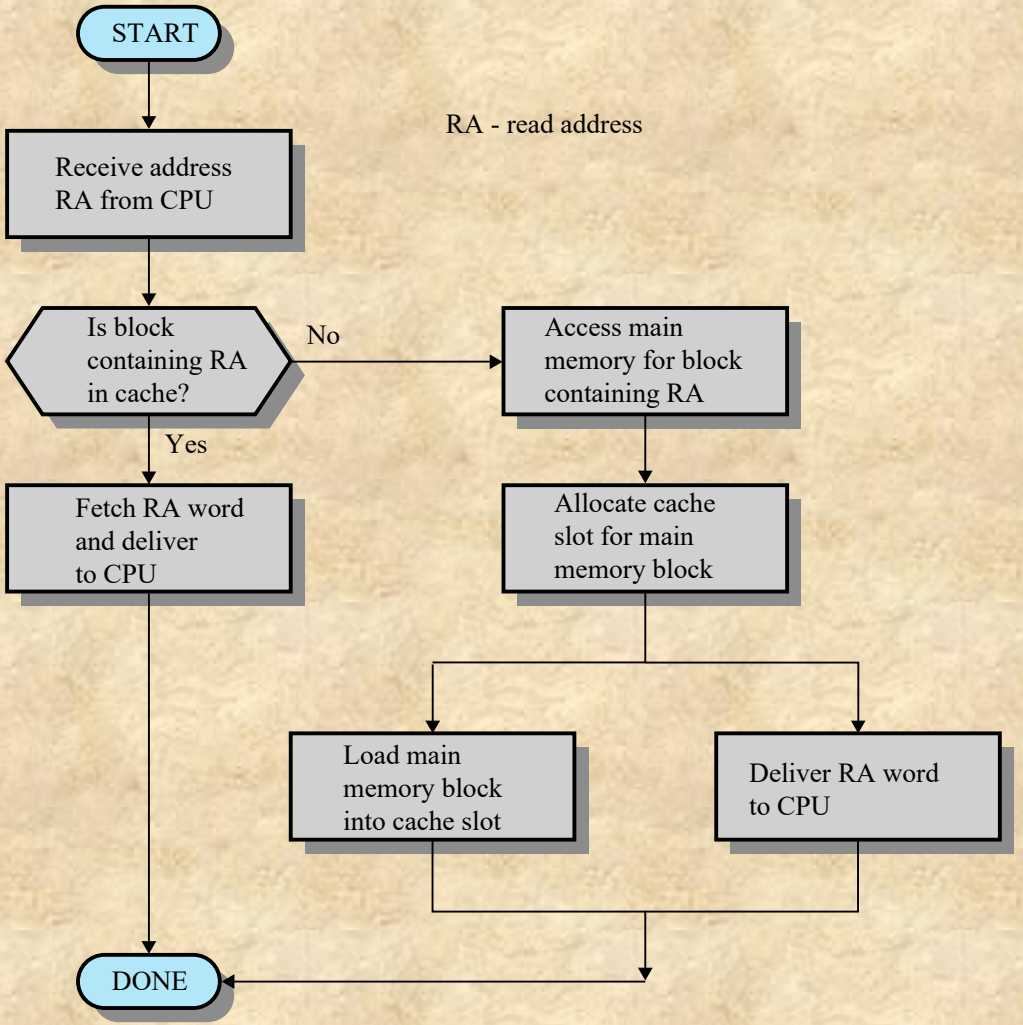
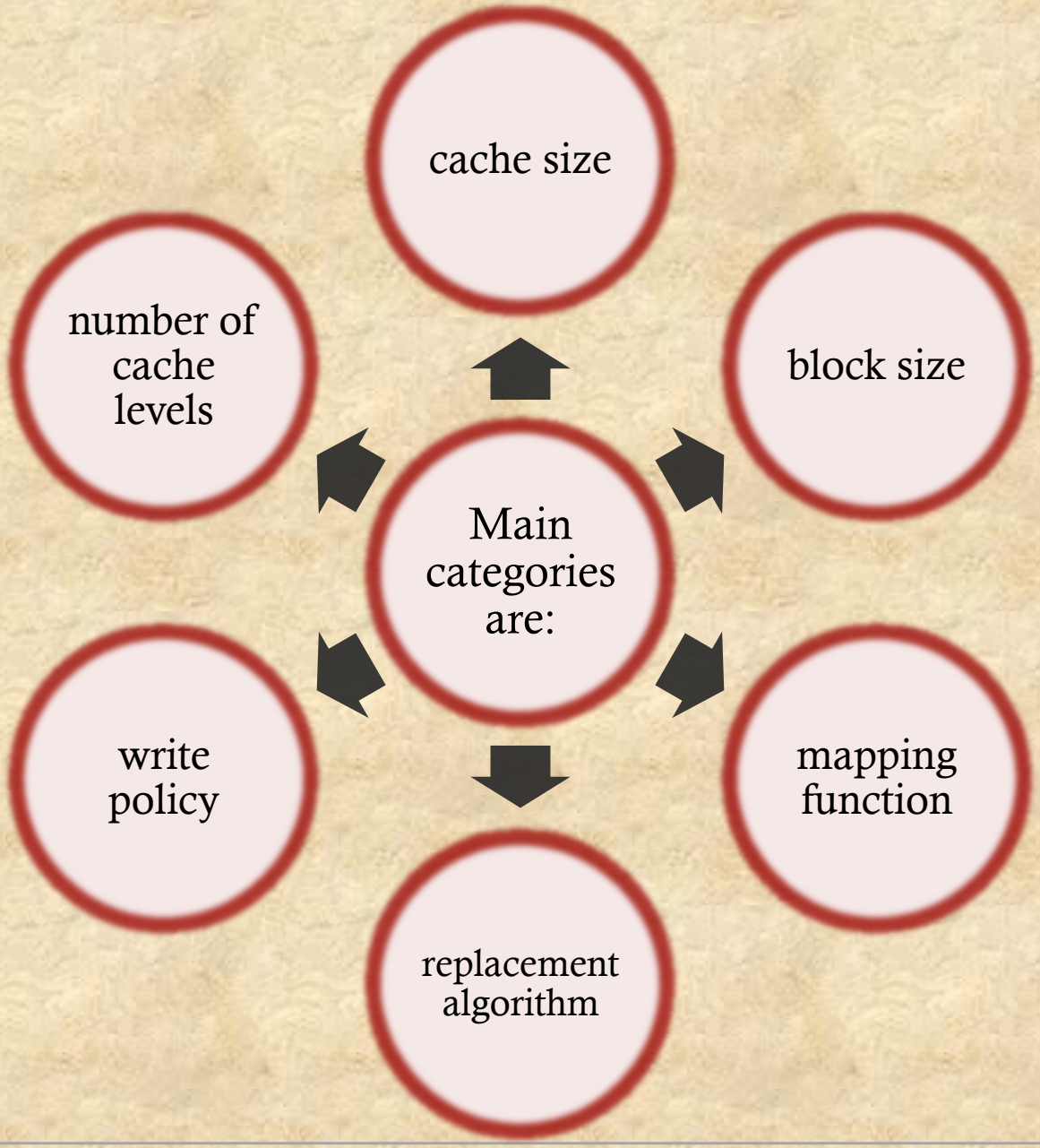


Figure 1.18 Cache Read Operation

Cache Design



Cache and Block Size

Cache Size

small caches have significant impact on performance

Block Size

the unit of data exchanged between cache and main memory

Mapping Function

* Determines which cache location the block will occupy

Two constraints affect design:

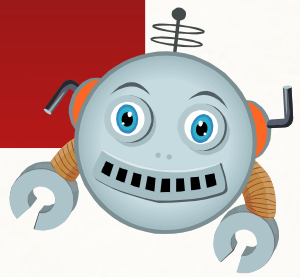
- when one block is read in, another may have to be replaced
- the more flexible the mapping function, the more complex is the circuitry required to search the cache

Replacement Algorithm

- Least Recently Used (LRU) Algorithm
  - effective strategy is to replace a block that has been in the cache the longest with no references to it
  - hardware mechanisms are needed to identify the least recently used block
    - chooses which block to replace when a new block is to be loaded into the cache

Write Policy

Dictates when the memory write operation takes place



- can occur every time the block is updated
- can occur when the block is replaced
  - minimizes write operations
  - leaves main memory in an obsolete state

I/O Techniques

* When the processor encounters an instruction relating to I/O, it executes that instruction by issuing a command to the appropriate I/O module

Three techniques are possible for I/O operations:

- Programmed I/O
- Interrupt-Driven I/O
- Direct Memory Access (DMA)

Programmed I/O

- The I/O module performs the requested action then sets the appropriate bits in the I/O status register
- The processor periodically checks the status of the I/O module until it determines the instruction is complete
- With programmed I/O the performance level of the entire system is severely degraded

Interrupt-Driven I/O

Processor issues an I/O command to a module and then goes on to do some other useful work

The I/O module will then interrupt the processor to request service when it is ready to exchange data with the processor

The processor executes the data transfer and then resumes its former processing

More efficient than Programmed I/O but still requires active intervention of the processor to transfer data between memory and an I/O module

Interrupt-Driven I/O Drawbacks



- Transfer rate is limited by the speed with which the processor can test and service a device
- The processor is tied up in managing an I/O transfer
  - A number of instructions must be executed for each I/O transfer

Direct Memory Access (DMA)

* Performed by a separate module on the system bus or incorporated into an I/O module

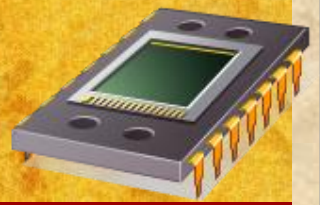
When the processor wishes to read or write data it issues a command to the DMA module containing:

- whether a read or write is requested
- the address of the I/O device involved
- the starting location in memory to read/write
- the number of words to be read/written

Direct Memory Access

- Transfers the entire block of data directly to and from memory without going through the processor
  - processor is involved only at the beginning and end of the transfer
  - processor executes more slowly during a transfer when processor access to the bus is required
- More efficient than interrupt-driven or programmed I/O

Symmetric Multiprocessors (SMP)



- A stand-alone computer system with the following characteristics:
  - two or more similar processors of comparable capability
  - processors share the same main memory and are interconnected by a bus or other internal connection scheme
  - processors share access to I/O devices
  - all processors can perform the same functions
  - the system is controlled by an integrated operating system that provides interaction between processors and their programs at the job, task, file, and data element levels

SMP Advantages

Performance

- a system with multiple processors will yield greater performance if work can be done in parallel

Scaling

- vendors can offer a range of products with different price and performance characteristics

Availability

- the failure of a single processor does not halt the machine

Incremental Growth

- an additional processor can be added to enhance performance

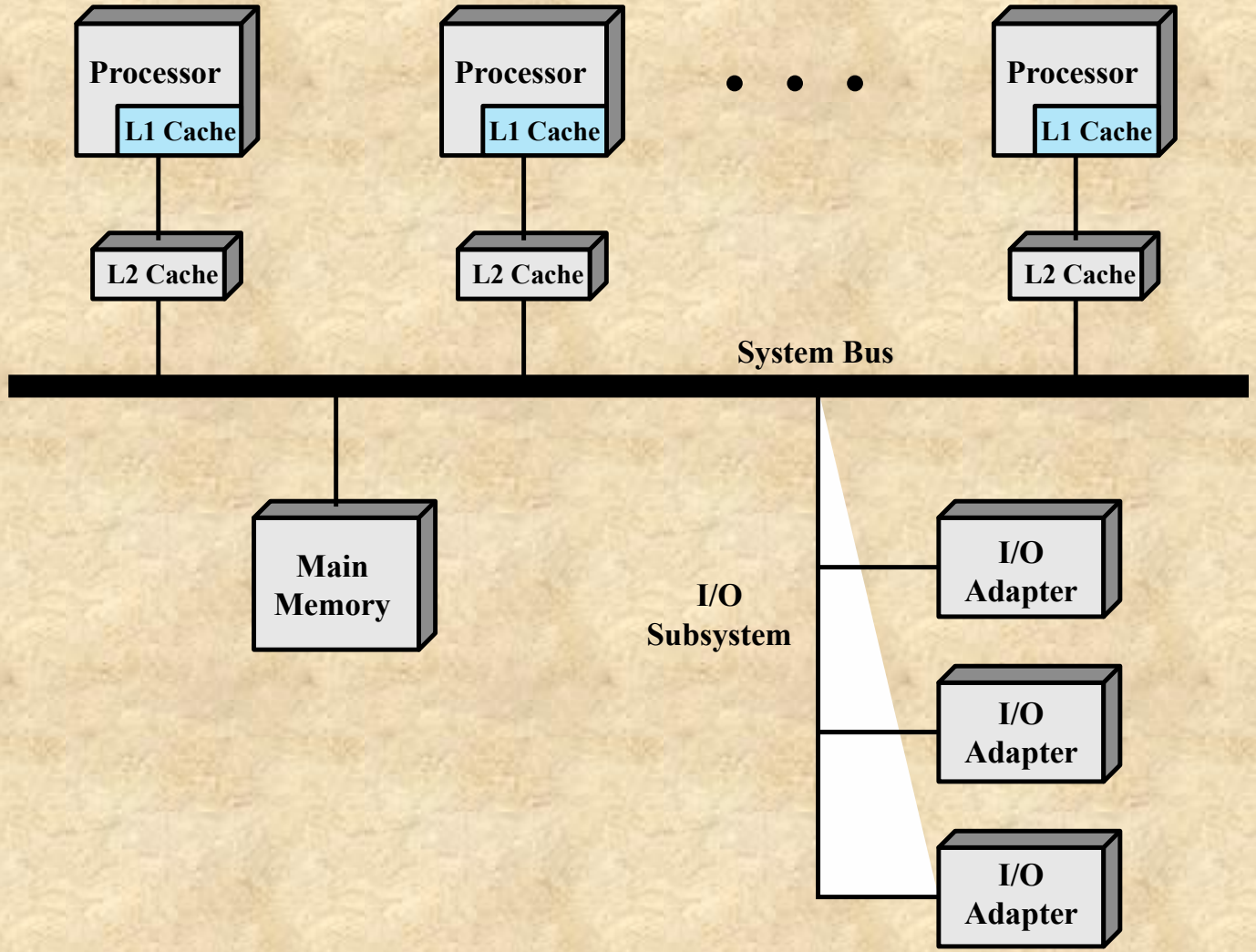
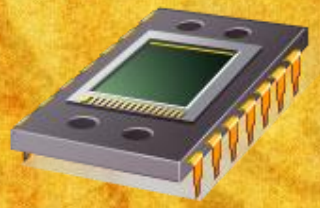


Figure 1.19 Symmetric Multiprocessor Organization

Multicore Computer



- Also known as a chip multiprocessor
- Combines two or more processors (cores) on a single piece of silicon (die)
  - each core consists of all of the components of an independent processor
- In addition, multicore chips also include L2 cache and in some cases L3 cache

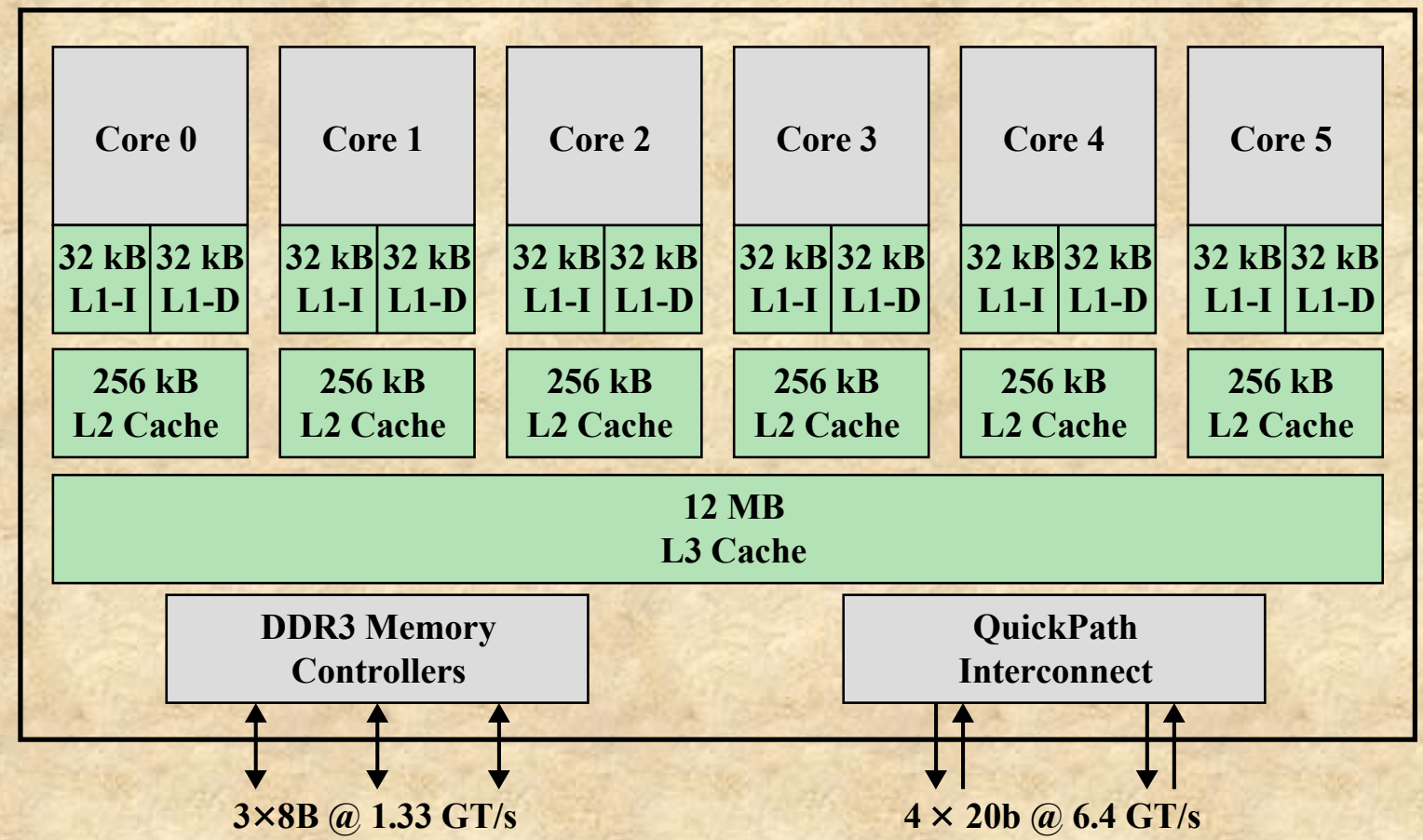
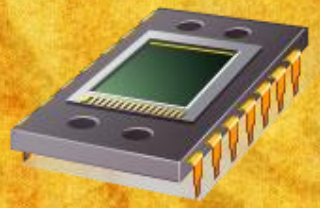


Figure 1.20 Intel Core i7-990X Block Diagram

Summary

- Basic Elements
- Evolution of the microprocessor
- Instruction execution
- Interrupts
  - Interrupts and the instruction cycle
  - Interrupt processing
  - Multiple interrupts
- The memory hierarchy
- Cache memory
  - Motivation
  - Cache principles
  - Cache design
- Direct memory access
- Multiprocessor and multicore organization
  - Symmetric multiprocessors
  - Multicore computers

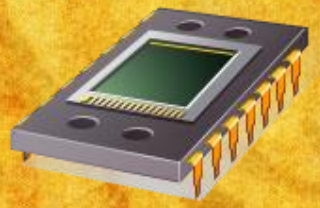
Quiz



- In operating system, each process has its own?
  a) address space and global variables
  b) open files
  c) pending alarms, signals and signal handlers
  d) all the above

Answer: d
Explanation: In Operating Systems, each process has its own address space which contains code, data, stack and heap segments or sections. Each process also has a list of files which is opened by the process as well as all pending alarms, signals and various signal handlers.

Quiz

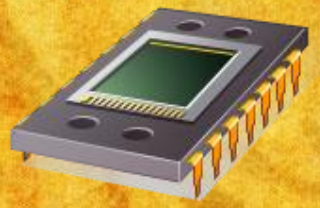


- A process can be terminated due to?
  a) normal exit
  b) fatal error
  c) killed by another process
  d) all the mentioned

- **Answer : d**

Explanation: A process can be terminated normally by completing its task or because of fatal error or killed by another process or forcefully killed by a user. When the process completes its task without any error then it exits normally.

Quiz

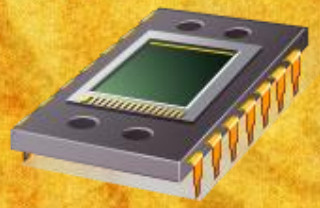


- The transfer between CPU and Cache is?
  a) Block transfer
  b) Word transfer
  c) Set transfer
  d) Associative transfer

- **Answer : b**

Explanation: The transfer is a word transfer. In the memory subsystem, word is transferred over the memory data bus, and it typically has a width of a word or half-word.

Quiz

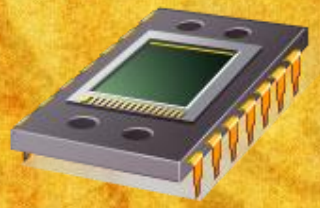


- The signal sent to the device from the processor to the device after receiving an interrupt is?
  a) Interrupt-acknowledge
  b) Return signal
  c) Service signal
  d) Permission signal

- **Answer : a**

Explanation: The Processor upon receiving the interrupt should let the device know that its request is received.

Quiz

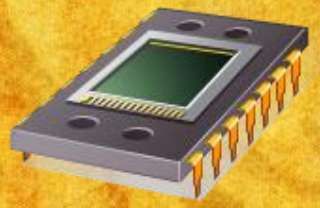


- Whenever is the data found in the cache memory it is called as?
  a) HIT
  b) MISS
  c) FOUND
  d) ERROR

- **Answer : a**

Explanation: Whenever the data is found in the cache memory, it is called as Cache HIT. CPU first checks in the cache memory since it is closest to the CPU.

Quiz



- What is an interrupt in an operating system?

  a. A software program

  b. A hardware signal

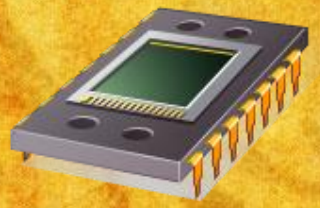
  c. A system call

  d. A user input

- **Answer : b**

Explanation: An interrupt in an operating system is a hardware signal generated by a device

Quiz

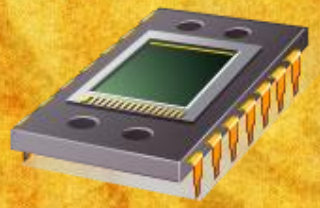


- The time between the receiver of an interrupt and its service is?
  a) Interrupt delay
  b) Interrupt latency
  c) Cycle time
  d) Switching time

- **Answer : b**

Explanation: : The delay in servicing of an interrupt happens due to the time is taken for contact switch to take place.

Quiz

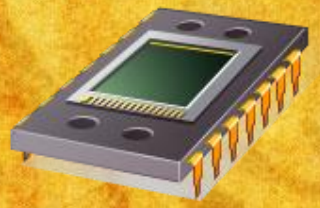


- An interrupt that can be temporarily ignored is?
  a) Vectored interrupt
  b) Non-maskable interrupt
  c) Maskable interrupt
  d) High priority interrupt

- **Answer : c**

Explanation: The maskable interrupts are usually low priority interrupts which can be ignored if a higher priority process is being executed.

Quiz

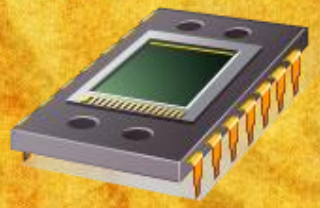


- Interrupts form an important part of _____ systems.
  a) Batch processing
  b) Multitasking
  c) Real-time processing
  d) Multi-user

- **Answer : c**

- Explanation: This forms an important part of the Real time system since if a process arrives with greater priority then it raises an interrupt and the other process is stopped and the interrupt will be serviced.

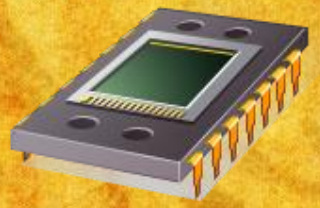
Quiz



- How can the processor ignore other interrupts when it is servicing one?
  a) By turning off the interrupt request line
  b) By disabling the devices from sending the interrupts
  c) BY using edge-triggered request lines
  d) All of the mentioned

- **Answer : d**

Quiz

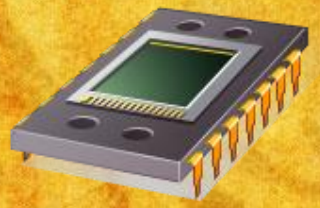


- While CPU is executing a program, an interrupt exists then it
  a) follows the next instruction in the program
  b) jumps to instruction in other registers
  c) breaks the normal sequence of execution of instructions
  d) stops executing the program

- **Answer : c**

Explanation: An interrupt function is to break the sequence of operation.

Quiz

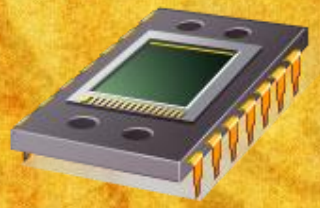


- CPU has two modes privileged and non-privileged. In order to change the mode from privileged to non-privileged.
  a) A hardware interrupt is needed
  b) A software interrupt is needed
  c) Either hardware or software interrupt is needed
  d) A non-privileged instruction (which does not generate an interrupt)is needed

Answer : b

Explanation: A software interrupt by some program which needs some CPU service, at that time the two modes can be interchanged.

Quiz

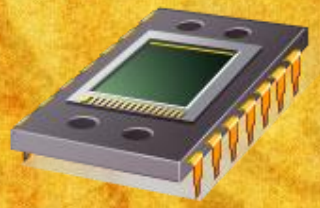


- While executing the main program, if two or more interrupts occur, then the sequence of appearance of interrupts is called
  a) multi-interrupt
  b) nested interrupt
  c) interrupt within interrupt
  d) nested interrupt and interrupt within interrupt

- **Answer : d**

Explanation: If an interrupt occurs while executing a program, and the processor is executing the interrupt, if one more interrupt occurs again, then it is called a nested interrupt.

Quiz

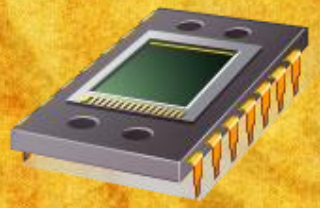


- If any interrupt request given to an input pin cannot be disabled by any means, then the input pin is called
  a) maskable interrupt
  b) nonmaskable interrupt
  c) maskable interrupt and nonmaskable interrupt
  d) none of the mentioned

- **Answer : b**

Explanation : A nonmaskable interrupt input pin is one which means that any interrupt request at NMI (nonmaskable interrupt) input cannot be masked or disabled by any means.

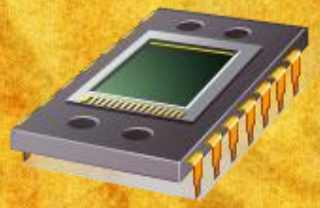
Quiz



- In Unix, Which system call creates the new process?
  a) fork
  b) create
  c) new
  d) none of the mentioned

Answer : a
Explanation: In UNIX, a new process is created by fork() system call. fork() system call returns a process ID which is generally the process id of the child process created.

Quiz

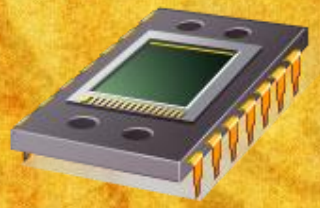


- The DMA transfer is initiated by?
  a) Processor
  b) The process being executed
  c) I/O devices
  d) OS

Answer : c

Explanation: The transfer can only be initiated by an instruction of a program being executed.

Quiz

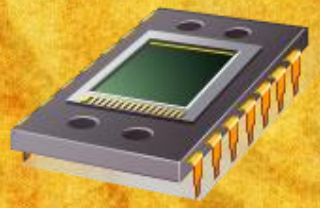


- When the process requests for a DMA transfer?
  a) Then the process is temporarily suspended
  b) The process continues execution
  c) Another process gets executed
  d) process is temporarily suspended & Another process gets executed

Answer : d

Explanation: The process requesting the transfer is paused and the operation is performed, meanwhile another process is run on the processor.

Quiz



- In an SMP system, how are tasks typically distributed among the multiple processors?

  a. Each processor runs a separate operating system

  b. Tasks are evenly distributed to all processors

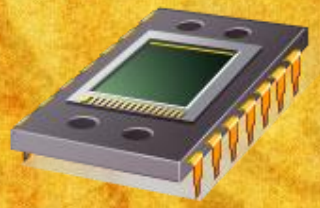
  c. Tasks are assigned based on priority

  d. Only one processor can execute tasks at a time

- **Answer : b**

Explanation: Tasks are evenly distributed to all processor

Quiz



- What is a key advantage of SMP systems?

  a. Reduced hardware complexity

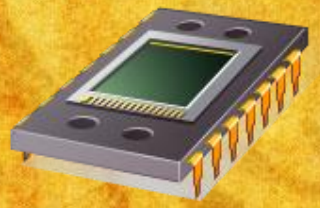
  b. Improved power efficiency

  c. Increased processing power and parallelism

  d. Simplicity in software development

Answer : c

Quiz



- Which of the following is a characteristic of SMP systems?

  a. Asymmetric memory access

  b. Exclusive use of single-core processors

  c. Single-threaded execution

  d. Load balancing among processors

Answer : d